



APURV SHASHVAT

B.E. Computer Science and Engineering
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PROFESSIONAL SUMMARY

Full-Stack Software Engineer with 2 years of experience building production-grade, AI-powered web applications using **Python (FastAPI, Django)**, **React / Next.js**, **PostgreSQL**. Currently building the backend and admin platform of an ed-tech product serving 50,000+ users, with hands-on ownership of REST API design, LLM/RAG pipelines, event-driven AWS workloads, database migrations and role-based access control.

EXPERIENCE

- **Zryth Solutions Pvt. Ltd.** Sept 2025 - present
Software Engineer – Oswaal Books(Ed-Tech)
 - **Built the FastAPI + PostgreSQL backend** for Oswaal360 AI, serving 50,000+ students across the React web app and Android app through a Node.js/Express BFF layer, keeping APIs backward-compatible across clients.
 - **Designed a RAG content pipeline** (LangChain, LangGraph, Gemini, Elasticsearch) that ingests academic books and auto-generates quizzes, mock tests, mind maps, flashcards, PPTs and AI podcasts.
 - **Moved heavy workloads to event-driven AWS infrastructure** – SQS queues, Lambda, ECS Fargate (podcast audio generation), S3 and RDS PostgreSQL – keeping API responses fast under load.
 - **Built an Automation tool** converting question-paper PDFs into mock tests using PyMuPDF, OCR and image-first LLM analysis with LaTeX rendering, cutting manual test-creation effort.
 - **Shipped student-facing features** – mock tests with leaderboards, weak-area based recommendation engine (background generation + polling), Exam Calendar and Analytics Hub – across web and React Native.
 - **Engineered Work Allotment** with per-assignment permission scoping, JWT authentication and phone OTP verification.
- **Keywords Studios** May 2025 - Aug 2025
Research Associate Engineer
 - **Trained Amazon's Agentic AI model- Nova** by identifying workflow errors, rewriting flawed prompts, and writing Chain of Thought reasoning for human-like task execution.
 - **Rubric-based assessment** to score AI behavior, flag hallucinations, and enhance interpretability for RLHF pipelines.
- **Nirmaan Organization** May 2024 - Mar 2025
Frontend Developer
 - **Improved website performance, achieving a 20%** faster page loading speed by implementing code minification, image optimization, and lazy loading techniques.
 - **Tested and debugged code** to ensure cross-browser compatibility and resolved 15% of identified issues, improving overall website reliability.

PROJECTS

- **Secure AI Vault - {Live link}** Mar 2024 - Apr 2024
Tools & technologies used : React, TypeScript, Node.js, Express, Firebase, REST API
 - Built a secure AI-powered vault to store and manage sensitive documents (credit/debit, Aadhaar, PAN cards), with AI-driven extraction of card details from uploaded or camera-captured images.

EDUCATION

- **VTU University , Bengaluru** 2024
B.E (CSE) CGPA/Percentage: 8.35

TECHNICAL SKILLS

Languages: Python, TypeScript, JavaScript, SQL, C++

Backend: FastAPI, Django, Node.js, Express.js, REST APIs, SQLAlchemy, JWT

Frontend: React.js, Next.js, Redux, TanStack Query, Tailwind CSS, React Native, HTML, CSS

Databases: PostgreSQL, Elasticsearch, Firebase

Cloud & DevOps: AWS (EC2, S3, RDS, Lambda, SQS, ECS Fargate), Docker, GitHub Actions, Linux, Git

ACHIEVEMENTS

- **Qualified to 2nd round of Flipkart Grid 5.0**, Solved 150+ problems on GFG, CodingNinja and HackerRank.
- **Research Paper**, Developed multilingual fake news detection system. (Published in Journal at JETIR2405D18).